

Hanif Bin Ariffin

Systems Engineer

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Github
LinkedIn
Personal Blog

TECHNICAL SKILLS

Languages

C++ | Rust | C | Go | Python | TypeScript | VHDL

Systems & Low Latency

Kernel Bypass (ef_vi) | SolarFlare | Lock-free structures | Shared Memory | HFT Infrastructure

Data & Infrastructure

Redis Internals | Market Data | Big Data (TB scale) | Linux/Unix | Docker

EDUCATION

BSc in Computer Engineering
University of Ottawa
Aug 2015 – Aug 2020
Canada

Systems Engineer specialized in high-performance computing and low-latency infrastructure. Proven track record in optimizing C++ applications at nanosecond scales, processing terabytes of market data, and contributing to core open-source infrastructure like Redis and Rust Coreutils.

WORK EXPERIENCE

Software Developer
Dec 2023 – Present
Confinex Technologies | *Pulau Pinang*

Developing and maintaining an entire trading infrastructure stack for high-frequency trading.

- Optimized C++ lock-free shared memory queue library for sub-microsecond IPC.
- Developed market data processing library with <400ns latency at P99.
- Built kernel bypass C++ applications using SolarFlare's ef_vi for direct NIC access.
- Engineered high-performance C++ bridges for Python JIT interaction with brokers.
- Automated processing of terabytes of market data across multiple global exchanges.

Senior Fullstack Developer
Feb 2022 – Jul 2022
Globelise | *Singapore*

Designed and implemented the backend for a global hiring management system.

- Led development of 3rd party payroll integrations for international markets.
- Implemented secure, scalable backend architecture for multi-region deployment.

Fullstack developer
Dec 2020 – Feb 2022
Setel (Petronas) | *Malaysia*

Maintained the loyalty and inventory systems for Malaysia's leading e-wallet.

- Scaled the store ordering system to handle nationwide self-checkout volume.
- Optimized inventory tracking logic to ensure real-time consistency.

OPEN SOURCE & CONTRIBUTIONS

Redis – *Contributor*

- Fixed critical crash in ZINTER involving SET and ZSET types.
- Optimized integer checks and suggested SDS memory layout improvements.

Rust Coreutils – *Contributor*

- Fully reimplemented 'tr' utility to match POSIX/GNU standards (100% test pass).
- Performance and stability improvements to 'ls' and 'more'.

Godot Engine – *Contributor*

- Resolved memory leaks and UI-related crashes in the core engine.

KEY PROJECTS

radish | High-performance Redis clone in Go.

- 90% of reference Redis performance using multithreaded architecture.

MIPS Processor | VHDL circuit with forwarding unit and branch protection.

- Implemented full instruction pipeline with hazard mitigation.